Newman-Penrose Formalism Calculations

Stephani et al., Equation (21.61_e1)

Metric

$$ds^2 = -\frac{x^2}{N^2}\mathrm{d}t\otimes\mathrm{d}t - \frac{n}{N^2}\mathrm{d}t\otimes\mathrm{d}\varphi + \frac{x^2}{(x^4+bx^2\log(x)+cx^2+n^2)N^2}\mathrm{d}x\otimes\mathrm{d}x - \frac{n}{N^2}\mathrm{d}\varphi\otimes\mathrm{d}t + \left(\frac{x^2+b\log(x)+c}{N^2}\right)\mathrm{d}\varphi\otimes\mathrm{d}\varphi + \frac{1}{N^2}\mathrm{d}z\otimes\mathrm{d}z$$

Coordinates

$$(x^\mu) = (t, x, \varphi, z)$$

Metric Functions

$$N(z) = N$$

$$G(x) = x^2 + b\log(x) + c + \frac{n^2}{x^2}$$

Null Tetrad

Covariant components

$$l = \left(\frac{\sqrt{2}x}{2\,N}\right)\,dt + \left(-\frac{\sqrt{2}\sqrt{x^2+b\log(x)+c+\frac{n^2}{x^2}}}{2\,N} + \frac{\sqrt{2}n}{2\,xN}\right)\,d\varphi$$

$$n = \left(\frac{\sqrt{2}x}{2\,N}\right)\,dt + \left(\frac{\sqrt{2}\sqrt{x^2+b\log(x)+c+\frac{n^2}{x^2}}}{2\,N} + \frac{\sqrt{2}n}{2\,xN}\right)\,d\varphi$$

$$m = \left(\frac{1}{\sqrt{2\,x^2+2\,b\log(x)+2\,c+\frac{2\,n^2}{x^2}}N}\right)\,dx + \left(\frac{i\,\sqrt{2}}{2\,N}\right)\,dz$$

$$\overline{m} = \left(\frac{1}{\sqrt{2\,x^2 + 2\,b\log\left(x\right) + 2\,c + \frac{2\,n^2}{x^2}}N}\right)\,dx + \left(-\frac{i\,\sqrt{2}}{2\,N}\right)\,dz$$

Contravariant components

$$l = \left(-\frac{\sqrt{2}x^4N}{2\left(x^5 + bx^3\log\left(x\right) + cx^3 + n^2x\right)} - \frac{\sqrt{2}bx^2N\log\left(x\right)}{2\left(x^5 + bx^3\log\left(x\right) + cx^3 + n^2x\right)} - \frac{\sqrt{2}cx^2N}{2\left(x^5 + bx^3\log\left(x\right) + cx^3 + n^2x\right)} - \frac{\sqrt{2}n^2N}{2\left(x^5 + bx^3\log\left(x\right) + cx^3 + n^2x\right)} + \frac{\sqrt{2}\sqrt{x^4 + bx^2\log\left(x\right) + cx^2 + n^2}nN}{2\left(x^5 + bx^3\log\left(x\right) + cx^3 + n^2x\right)}\right)\,dt + \left(-\frac{\sqrt{2}xN}{2\,\sqrt{x^4 + bx^2\log\left(x\right) + cx^2 + n^2}}\right)\,d\varphi$$

$$n = \left(-\frac{\sqrt{2}x^4N}{2\left(x^5 + bx^3\log\left(x\right) + cx^3 + n^2x\right)} - \frac{\sqrt{2}bx^2N\log\left(x\right)}{2\left(x^5 + bx^3\log\left(x\right) + cx^3 + n^2x\right)} - \frac{\sqrt{2}cx^2N}{2\left(x^5 + bx^3\log\left(x\right) + cx^3 + n^2x\right)} - \frac{\sqrt{2}n^2N}{2\left(x^5 + bx^3\log\left(x\right) + cx^3 + n^2x\right)} - \frac{\sqrt{2}\sqrt{x^4 + bx^2\log\left(x\right) + cx^2 + n^2}nN}{2\left(x^5 + bx^3\log\left(x\right) + cx^3 + n^2x\right)}\right)\,dt + \left(\frac{\sqrt{2}xN}{2\,\sqrt{x^4 + bx^2\log\left(x\right) + cx^2 + n^2}}\right)\,d\varphi$$

$$m = \left(\frac{\sqrt{2}\sqrt{x^4 + bx^2\log\left(x\right) + cx^2 + n^2}N}{2\,x}\right)\,dx + \left(\frac{1}{2}i\,\sqrt{2}N\right)\,dz$$

$$\overline{m} = \left(\frac{\sqrt{2}\sqrt{x^4 + bx^2\log\left(x\right) + cx^2 + n^2}N}{2\,x}\right)\,dx + \left(-\frac{1}{2}i\,\sqrt{2}N\right)\,dz$$

Directional Derivatives

$$D = -\frac{\sqrt{2}x^4N}{2\left(x^5 + bx^3\log\left(x\right) + cx^3 + n^2x\right)} - \frac{\sqrt{2}bx^2N\log\left(x\right)}{2\left(x^5 + bx^3\log\left(x\right) + cx^3 + n^2x\right)} - \frac{\sqrt{2}cx^2N}{2\left(x^5 + bx^3\log\left(x\right) + cx^3 + n^2x\right)} - \frac{\sqrt{2}n^2N}{2\left(x^5 + bx^3\log\left(x\right) + cx^3 + n^2x\right)} + \frac{\sqrt{2}\sqrt{x^4 + bx^2\log\left(x\right) + cx^2 + n^2}nN}{2\left(x^5 + bx^3\log\left(x\right) + cx^3 + n^2x\right)}\,\partial_t - \frac{\sqrt{2}xN}{2\,\sqrt{x^4 + bx^2\log\left(x\right) + cx^2 + n^2}}\,\partial_\varphi$$

$$\Delta = -\frac{\sqrt{2}x^4N}{2\left(x^5 + bx^3\log\left(x\right) + cx^3 + n^2x\right)} - \frac{\sqrt{2}bx^2N\log\left(x\right)}{2\left(x^5 + bx^3\log\left(x\right) + cx^3 + n^2x\right)} - \frac{\sqrt{2}cx^2N}{2\left(x^5 + bx^3\log\left(x\right) + cx^3 + n^2x\right)} - \frac{\sqrt{2}n^2N}{2\left(x^5 + bx^3\log\left(x\right) + cx^3 + n^2x\right)} - \frac{\sqrt{2}\sqrt{x^4 + bx^2\log\left(x\right) + cx^2 + n^2}nN}{2\left(x^5 + bx^3\log\left(x\right) + cx^3 + n^2x\right)}\,\partial_t + \frac{\sqrt{2}xN}{2\,\sqrt{x^4 + bx^2\log\left(x\right) + cx^2 + n^2}}\,\partial_\varphi$$

$$\delta = \frac{\sqrt{2}\sqrt{x^4 + bx^2 \log(x) + cx^2 + n^2}N}{2\,x}\,\partial_x + \frac{1}{2}i\,\sqrt{2}N\,\partial_z$$

$$\overline{\delta} = \frac{\sqrt{2}\sqrt{x^4 + bx^2 \log(x) + cx^2 + n^2}N}{2\,x}\,\partial_x - \frac{1}{2}i\,\sqrt{2}N\,\partial_z$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\kappa = \frac{\sqrt{2}nx^4N}{2\left(x^6 + bx^4\log(x) + cx^4 + n^2x^2\right)} + \frac{\sqrt{2}bnx^2N\log(x)}{2\left(x^6 + bx^4\log(x) + cx^4 + n^2x^2\right)} - \frac{\sqrt{2}\sqrt{x^4 + bx^2\log(x) + cx^2 + n^2}bx^2N\log(x)}{4\left(x^6 + bx^4\log(x) + cx^4 + n^2x^2\right)} + \frac{\sqrt{2}cnx^2N}{2\left(x^6 + bx^4\log(x) + cx^4 + n^2x^2\right)} + \frac{\sqrt{2}\sqrt{x^4 + bx^2\log(x) + cx^2 + n^2}bx^2N}{8\left(x^6 + bx^4\log(x) + cx^4 + n^2x^2\right)} - \frac{\sqrt{2}\sqrt{x^4 + bx^2\log(x) + cx^2 + n^2}cx^2N}{4\left(x^6 + bx^4\log(x) + cx^4 + n^2x^2\right)} + \frac{\sqrt{2}n^3N}{2\left(x^6 + bx^4\log(x) + cx^4 + n^2x^2\right)} - \frac{\sqrt{2}\sqrt{x^4 + bx^2\log(x) + cx^2 + n^2}n^2N}{2\left(x^6 + bx^4\log(x) + cx^4 + n^2x^2\right)}$$

$$\mu = 0$$

$$\lambda = 0$$

$$\nu = \frac{\sqrt{2}nx^4N}{2\left(x^6 + bx^4\log(x) + cx^4 + n^2x^2\right)} + \frac{\sqrt{2}bnx^2N\log(x)}{2\left(x^6 + bx^4\log(x) + cx^4 + n^2x^2\right)} + \frac{\sqrt{2}\sqrt{x^4 + bx^2\log(x) + cx^2 + n^2}bx^2N\log(x)}{4\left(x^6 + bx^4\log(x) + cx^4 + n^2x^2\right)} + \frac{\sqrt{2}cnx^2N}{2\left(x^6 + bx^4\log(x) + cx^4 + n^2x^2\right)} - \frac{\sqrt{2}\sqrt{x^4 + bx^2\log(x) + cx^2 + n^2}bx^2N}{8\left(x^6 + bx^4\log(x) + cx^4 + n^2x^2\right)} + \frac{\sqrt{2}\sqrt{x^4 + bx^2\log(x) + cx^2 + n^2}cx^2N}{4\left(x^6 + bx^4\log(x) + cx^4 + n^2x^2\right)} + \frac{\sqrt{2}n^3N}{2\left(x^6 + bx^4\log(x) + cx^4 + n^2x^2\right)} + \frac{\sqrt{2}\sqrt{x^4 + bx^2\log(x) + cx^2 + n^2}n^2N}{2\left(x^6 + bx^4\log(x) + cx^4 + n^2x^2\right)}$$

$$\alpha = -\frac{1}{4}i\,\sqrt{2}\frac{\partial}{\partial z}N + \frac{\sqrt{2}nN}{4\,x^2}$$

$$\beta = -\frac{1}{4}i\,\sqrt{2}\frac{\partial}{\partial z}N + \frac{\sqrt{2}nN}{4\,x^2}$$

$$\tau = \frac{i\sqrt{2}x^4 \frac{\partial}{\partial z}N}{2\left(x^4 + bx^2\log(x) + cx^2 + n^2\right)} + \frac{i\sqrt{2}bx^2\log(x)\frac{\partial}{\partial z}N}{2\left(x^4 + bx^2\log(x) + cx^2 + n^2\right)} + \frac{i\sqrt{2}cx^2 \frac{\partial}{\partial z}N}{2\left(x^4 + bx^2\log(x) + cx^2 + n^2\right)} - \frac{\sqrt{2}x^2N}{2\sqrt{x^4 + bx^2\log(x) + cx^2 + n^2}}$$
$$- \frac{\sqrt{2}bN\log(x)}{4\sqrt{x^4 + bx^2\log(x) + cx^2 + n^2}} + \frac{i\sqrt{2}n^2\frac{\partial}{\partial z}N}{2\left(x^4 + bx^2\log(x) + cx^2 + n^2\right)} - \frac{\sqrt{2}bN}{8\sqrt{x^4 + bx^2\log(x) + cx^2 + n^2}} - \frac{\sqrt{2}cN}{4\sqrt{x^4 + bx^2\log(x) + cx^2 + n^2}}$$

$$\pi = \frac{i\sqrt{2}x^4 \frac{\partial}{\partial z}N}{2\left(x^4 + bx^2\log(x) + cx^2 + n^2\right)} + \frac{i\sqrt{2}bx^2\log(x)\frac{\partial}{\partial z}N}{2\left(x^4 + bx^2\log(x) + cx^2 + n^2\right)} + \frac{i\sqrt{2}cx^2 \frac{\partial}{\partial z}N}{2\left(x^4 + bx^2\log(x) + cx^2 + n^2\right)} + \frac{\sqrt{2}x^2N}{2\sqrt{x^4 + bx^2\log(x) + cx^2 + n^2}}$$
$$+ \frac{\sqrt{2}bN\log(x)}{4\sqrt{x^4 + bx^2\log(x) + cx^2 + n^2}} + \frac{i\sqrt{2}n^2\frac{\partial}{\partial z}N}{2\left(x^4 + bx^2\log(x) + cx^2 + n^2\right)} + \frac{\sqrt{2}bN}{8\sqrt{x^4 + bx^2\log(x) + cx^2 + n^2}} + \frac{\sqrt{2}cN}{4\sqrt{x^4 + bx^2\log(x) + cx^2 + n^2}}$$

$$\epsilon = 0$$

$$\gamma = 0$$

Ricci Tensor Components

$$\Phi_{00} = \frac{bN^2}{4\,x^2}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = -\frac{1}{2}\,N^2 - \frac{1}{2}\,N\frac{\partial^2}{(\partial z)^2}N$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{1}{4}\,N^2 + \frac{1}{4}\,N\frac{\partial^2}{(\partial z)^2}N + \frac{bN^2}{8\,x^2}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = -\frac{1}{2}\,N^2 - \frac{1}{2}\,N\frac{\partial^2}{(\partial z)^2}N$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = \frac{bN^2}{4\,x^2}$$

$$\Lambda = -\frac{1}{4}\,N^2 - \frac{1}{2}\,\frac{\partial}{\partial z}N^2 + \frac{1}{4}\,N\frac{\partial^2}{(\partial z)^2}N - \frac{bN^2}{24\,x^2}$$

Weyl Tensor Components

$$\Psi_0 = \frac{bN^2}{4\,x^2}$$

$$\Psi_1 = 0$$

$$\Psi_2 = \frac{bN^2}{12\,x^2}$$

$$\Psi_3 = 0$$

$$\Psi_4 = \frac{bN^2}{4\,x^2}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "D"